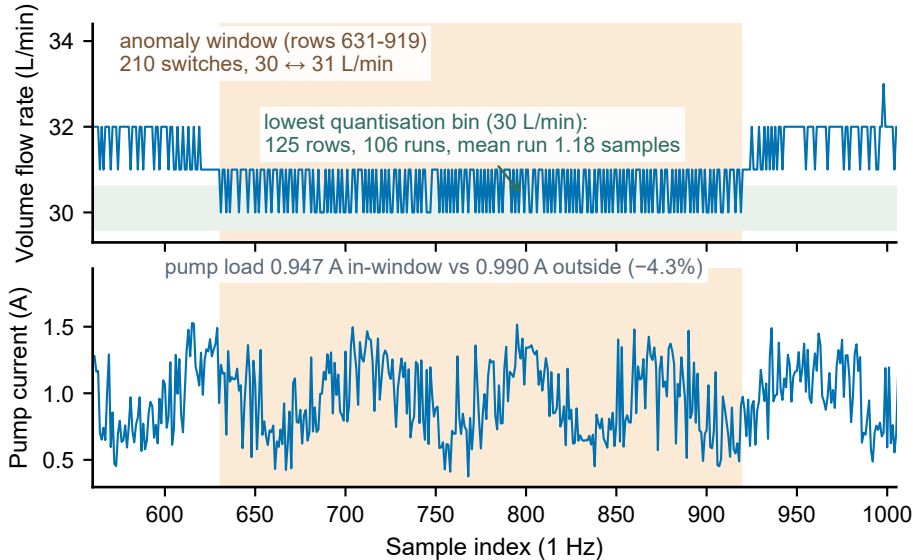




Evidence chain: L1 recorded signals (flow hunts in the two lowest bins; no vibration/thermal shift; valve-position channel absent) → L3 statistics (window mean 30.62 vs 31.73 L/min, -3.5%; 39.4% residence at lowest quantised level vs 0.0% outside) → L5 mechanism (downstream resistance ↑ forces the pump down its curve) → H1 valve/downstream restriction vs H2 range-floor meter artifact → verdict: capped COMPETING_SET 0.65 — named missing channel: valve-position/command feedback